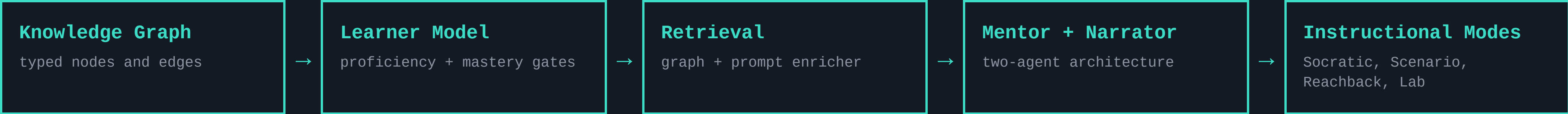


# TeachMe — Retrieve for the learner, not just the query.

TeachMe is a domain-agnostic platform for training users to operate complex, safety-critical equipment. It combines educational frameworks, agent-guided instruction, and retrieval from an expert-reviewed knowledge graph to provide grounded guidance, adaptive practice, and mastery-based progression.

*A technician learning the Optomec HD2 — or a mariner learning the rules of the road — needs answers at the right level of complexity, with a guarantee that the system will never hallucinate a safety value. If a technician internalizes even one incorrect parameter during training, they may leave more dangerous than when they started.*

## Architecture Overview



## What’s Working Now

NEW: WE NOW MEASURE WHETHER THE TUTOR ACTUALLY USES ITS CORPUS (COMPANION POSTER)

01

**Corpus-bound Narrator**  
Reports machine behavior verbatim from reviewed graph nodes; cannot generate from model weights.  
  
→ so a learner can’t internalize a hallucinated safety value; a silent corpus shows the gap, never fabricates it.

02

**Typed causal graph**  
100+ nodes, six fault domains: Symptom → FailureMode → CorrectiveAction with SafetyHazard co-retrieved along the chain.  
→ so safety context travels with the diagnosis instead of being a separate lookup a flat store can miss.

03

**Dual-axis mastery gates**  
Declarative proficiency (Socratic probes) must clear before procedural practice; gates are transparent and auditable.  
  
→ so an instructor can point at a score and say exactly why the system advanced or held a learner.

04

**Four modes, one progression**  
Socratic Practice → Scenario Mode (deterministic fault injection) → Reachback Lookup → Retrieval Lab.  
  
→ so training load is sequenced and instructors inspect what retrieval will surface before a learner sees it.

## Five-Level Expertise Continuum



## One Engine, Three Domains

**EDDIE · Aerosol Jet Printing — flagship**  
Optomec HD2 operator training; 100+ nodes, six fault domains, hazards co-retrieved.

**Roadside Tire Change**  
authored as graph + corpus alone — inherits probes, scenarios, gates, consequences.

**COLREG Collision Avoidance**  
rules of the road + interactive simulator: real kinematics, Collision Risk Index, per-rule scoring.

→ so a new domain is an authoring task, not a software project.

### IF THIS WORKS

A technician reaches independent, unsupervised operation faster and with fewer escalations — because training builds far-transfer diagnostic capability, not recall.

### TARGET

far-transfer performance on novel fault combinations vs. direct instruction — not yet measured on real learners; the Workflow Demo exists to de-risk this bet first.

## HOW WE KEEP OURSELVES HONEST

*“In safety-critical training, ‘I don’t know, but here’s the SOP for this state’ beats a confident wrong answer.”*

| SIGNAL                  | WHAT IT CAN CLAIM                                                                                                                            | TRUST         | CAN’T RULE OUT                                                                                        |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|---------------|-------------------------------------------------------------------------------------------------------|
| Corpus-bound Narrator   | Eliminates hallucinated machine behavior by construction — every observable traces to a reviewed node.                                       | <b>strong</b> | states outside the authored fault domains aren’t narratable (a coverage gap, made visible not hidden) |
| Corpus necessity audit  | We test whether answers actually depend on the corpus — a calibrated screen, verified on five different base models, before any human trial. | <b>strong</b> | evidence is from simulation — it audits the tutor’s grounding, not how well humans learn              |
| Dual-axis learner model | Transparent, auditable tutor decisions, inspectable in the Retrieval Lab.                                                                    | <b>strong</b> | deliberately simpler than probabilistic knowledge tracing — may miss subtler learner states           |
| Far-transfer outcome    | PBL design targets far transfer to novel fault combinations.                                                                                 | <b>weak</b>   | whether the simulated-learner loop predicts real technician outcomes                                  |

**LEFT OUT ON PURPOSE** free-running text generation. The Narrator can only quote the reviewed corpus — so no prompt, and no jailbreak, can make it invent machine behavior.

## THE QUESTIONS WE’RE CHASING

### How do we effectively use AI in high-consequence environments?

- Q1 Do learners trained on injected faults diagnose novel fault combinations better than those taught by direct instruction?
- Q2 Does passing a mastery gate predict safe, unsupervised operation on the real equipment?
- Q3 How far can the learning sciences take operational readiness in high-consequence work?



github.com/wrgr/socratic-scenarios

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